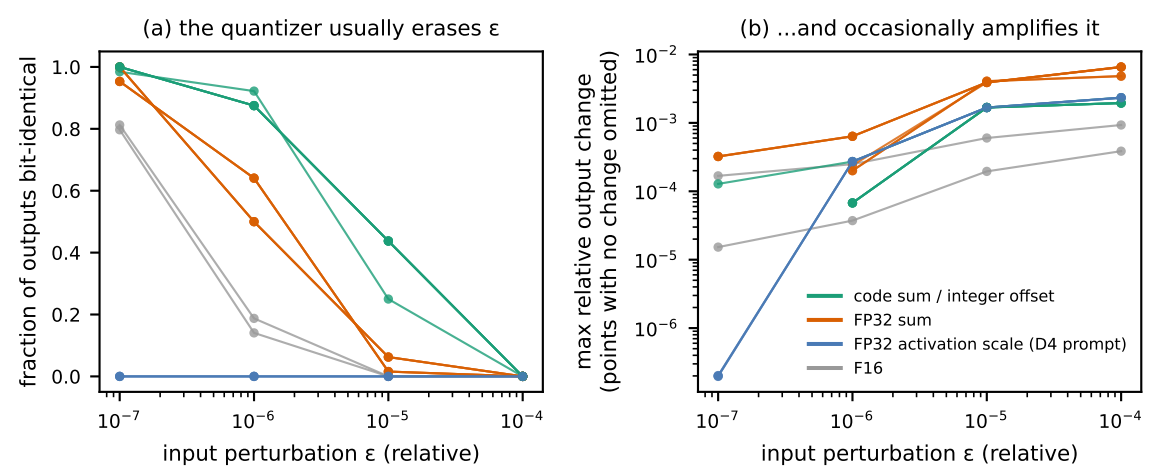
(a) the quantizer usually erases ε
fraction of outputs bit-identical
1.0
0.8
0.6
0.4
0.2
0.0
10^{-7}
10^{-6}
10^{-5}
10^{-4}
input perturbation ε (relative)
(b) ...and occasionally amplifies it
max relative output change
(points with no change omitted)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
code sum / integer offset
FP32 sum
FP32 activation scale (D4 prompt)
F16
10^{-7}
10^{-6}
10^{-5}
10^{-4}
input perturbation ε (relative)